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#include<stdio.h>
#include<math.h>
#define QSIZE 4
#define TRUE 1
#define FALSE 0

struct Queue
{
int rear, front;
int item[QSIZE];
};

struct Queue q;

void insert(int);
int delete();

int empty();

void main()
{
char ch;
int c,x,i;
q.rear=-1;
q.front=0;
do
{
printf("MENU\n");
printf("1. INSERT \n 2. DELETE\n 3. Display\n ");
scanf("%d",&c);
switch(c)
{
case 1: printf("enter the no to be inserted\n");
scanf("%d",&x);
insert(x);
break;
case 2: x=delete();
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        if(x!=-1)
            printf("deleted element is %d",x);
        break;
case 3: printf("elements are\n");
        for(i=q.rear;i>=q.front;i--)
        {
            printf("%d ", q.item[i]);
        }
        break;
    }
    printf("do you wish to continue\n");
    scanf(" %c",&ch);
}while(ch!='n');
}

```

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void insert(int x)
{
    if(q.rear==QSIZE-1)
    {
        printf("Queue overflow\n");
    }
    else
    {q.item[++q.rear]=x;
    }
}

```

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int delete()
{
    if(empty())
    {
        printf("Queue underflow\n");
        return(-1);
    }
    return(q.item[q.front++]);
}

```

```
int empty()
{
if (q.rear<q.front)
return TRUE;
else
return FALSE;
}
```